



CAL

Central Analytical Laboratories

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CERTIFICATE OF ANALYSIS

January 10, 2020

Client: Standard Process
1200 W. Royal Lee Drive
Palmyra, WI 53156 United States

COMMODITY: Hemp Seed and Stalk Oil

LOT NUMBER: L101818116

QA18L: Product Specification Review

1. The hemp seed and stalk oil (L101818116) meets the requirements for pesticides as defined by USP 561.
2. The hemp seed and stalk oil (L101818116) meets the requirements for heavy metals as defined by USP 2232.
3. The hemp seed and stalk oil (L101818116) meets the requirements for Microbiological levels as defined by USP 2023 (Botanical Extracts).

Victoria Siegel, Analytical Service Manager

ECAL ID: 19-12130024

A2LA ISO/IEC 17025:2005
Best Aquaculture Practices
Food and Drug Administration

American Oil Chemists Society
Grain and Feed Trade Association
Federation of Oils, Seed, and Fats Associations, Ltd.

Japanese Ministry of Health and Welfare
Association of Official Analytical Chemists
United States Department of Agriculture

CERTIFICATE OF ANALYSIS

**ALKEMIST LABS**

12661 HOOVER STREET GARDEN GROVE, CA 92841 | P. 714-754-4372 | F. 714-668-9972 | WWW.ALKEMIST.COM

Report Issued To: Standard Process, Inc
1200 West Royal Lee Drive
Palmyra WI 53156**Sample Name:** Hemp Stalk Oil
Description: Oil; Dark green liquid
Lot #: L101818116
AL #: JAR30518SP1_1
Analysis ID: 111985
Received: 11/01/18

Determination of Cannabinoids Content by UPLC

Ret. Time (min)	Compound Name	Prep	Amount (%)	Specification	Result
7.1	Δ9-Tetrahydrocannabinidiol (THC)	Prep 1, Inj 1	0.035	N/A	N/A
		Prep 1, Inj 2	0.035	N/A	N/A
		Prep 2, Inj 1	0.035	N/A	N/A
		Prep 2, Inj 2	0.039	N/A	N/A
		Average	0.036	<0.2%	Pass

Chromatographic Conditions:

Method: ATM-815-0229
Column: AP213 Xbridge BEH C18 2.5µm (3.0 x 100 mm)
Temperature: 30°C
Flow Rate: 0.4 mL/min
Injection Volume: 1 µL
UV Detection: 210 nm, 228
Mobile Phase: 0.1% Formic Acid in 10% Acetonitrile
0.1% Formic Acid in 90% Acetonitrile
HPLC Instrument: UPLC_1

Sample Preparation:

Transferred approximately 125 mg into 25 mL volumetric flask. Filled to volume with Methanol. Sonicated 15 minutes at room temperature. Filtered through 0.2 µm PTFE syringe into HPLC vial for analysis.

Report Summary:

Conclusion: This "Hemp Stalk Oil" test sample contains 0.04% THC on the as is basis.
OOS Reference: N/A
Notebook Reference: LC125 p. 120
Revision 1: Removed chromatogram

Analysis Date : 12/04/18
Rev 1: 12/18/19 CG

Analyzed By: C Gray

Authorized By: Kirtal Chopra,
Analytical Chemistry Supervisor



Kirtal Chopra

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CERTIFICATE OF ANALYSIS

January 10, 2020

Client: Standard Process
1200 W. Royal Lee Drive
Palmyra, WI 53156 United States

COMMODITY: Hemp Seed and Stalk Oil

LOT NUMBER: PO000065134-1-1

QA18L: Product Specification Review

1. The hemp seed and stalk oil (PO000065134-1-1) meets the requirements for pesticides as defined by USP 561.
2. The hemp seed and stalk oil (PO000065134-1-1) meets the requirements for heavy metals as defined by USP 2232.
3. The hemp seed and stalk oil (PO000065134-1-1) meets the requirements for Microbiological levels as defined by USP 223 (Botanical Extracts).

Victoria Siegel, Analytical Service Manager

ECAL ID: 19-12130025

A2LA ISO/IEC 17025:2005
Best Aquaculture Practices
Food and Drug Administration

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Grain and Feed Trade Association
Federation of Oils, Seed, and Fats Associations, Ltd.

Japanese Ministry of Health and Welfare
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CERTIFICATE OF ANALYSIS



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Report Issued To: Standard Process, Inc
1200 West Royal Lee Drive
Palmyra WI 53156

Sample Name: Hemp Stalk Oil
Description: Oil; Thick dark brown liquid
Lot #: PO000065134-1-1
AL #: 19135RFW_1
Analysis ID: 119594
Received: 05/15/19

Determination of Cannabinoids Content by UPLC

Ret. Time (min)	Compound Name	Prep 1 (%)	Prep 2 (%)	Average (%)	Spec	Result
7.0	Δ9 - Tetrahydrocannabinol (THC)	0.186	0.185	0.185	< 0.2%	Pass

Chromatographic Conditions:

Method: ATM-815-0229
Column: AP213 Xbridge BEH C18 2.5µm (3.0 x 100 mm)
Temperature: 30°C
Flow Rate: 0.4 mL/min
Injection Volume: 1 µL
UV Detection: 210 nm, 228
Mobile Phase: 0.1% Formic Acid in 10% Acetonitrile
0.1% Formic Acid in 90% Acetonitrile
HPLC Instrument: UPLC_1

Sample Preparation:

Transferred approximately 125 mg into 25 mL volumetric flask. Filled to volume with Methanol. Sonicated 15 minutes at room temperature. Filtered through 0.2 µm PTFE syringe filter into HPLC vial for analysis.

Report Summary:

Conclusion: This "Hemp Stalk Oil" test sample contains 0.2% THC on the as-is basis.
OOS Reference: N/A
Notebook Reference: LC113 p. 163
Revision 1: Removed chromatogram

Analysis Date : 05/20/19
Rev 1: 12/18/19 CG

Analyzed By: C Gray

Authorized By: Kirtal Chopra,
Analytical Chemistry Supervisor

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 Kirtal Chopra
Analytical Chemistry Supervisor

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